

ULC302 : INTRODUCTION TO NETWORK THEORY

L	T	P	Cr
3	1	0	3.5

Course Objectives: To make the students understand the concepts of graph theory, two port networks, filter design, attenuators, oscillator and network synthesis.

Syllabus

Graph Theory: Graph, Tree and link branches, Network matrices and their relations, Choice of linearly independent network variables, Topological equations for loop current and for nodal voltage, Duality.

Network Theorems: Superposition Theorem, Thevenin's theorem, Norton's theorem, and Maximum power transfer theorem as applied to A.C. circuits, Tellegen's theorem and their applications.

Two Port Networks: Two port network description in terms of open circuits impedance, Short circuit admittance, Hybrid and inverse hybrid, ABCD and inverse ABCD parameters, Inter-connection of two port network, Indefinite admittance matrix and its applications.

Network Functions: Concepts of complex frequency, Transform impedance, Networks function of one port and two port network, concepts of poles and zeros, property of driving point and transfer function.

Passive Network Synthesis: Introduction, Positive Real Functions: Definition, Necessary and sufficient conditions for a function to be positive real, Synthesis vs. analysis, Elements of circuit synthesis, Foster and Cauer forms of LC Networks, Synthesis of RC and RL networks.

Filters and Attenuators: Classification of filters, Analysis of a prototype low pass, High pass, Band pass, Band stop and Mderived filter, Attenuation, Types of attenuators: symmetrical and asymmetrical.

Active Filters: Introduction to Active filters, first and second order low pass Butterworth filter, First and second order high pass Butterworth filter, Band pass filter.

Course Learning Outcomes (CLOs) /Course Outcomes (COs):

On completion of this course, the students will be able to:

1. Apply graph theory concept to solve electrical networks
2. Analyse the behaviour of two port networks
3. Apply various laws and theorems to solve electric networks
4. Realize network function of single and two port networks
5. Realize synthesis of different passive networks
6. Design different filter and attenuator configurations

Text Books:

1. Hayt, W., Engineering Circuit Analysis, Tata McGraw-Hill (2006).
2. Hussain, A., Networks and Systems, CBS Publications (2004).
3. Valkenberg, Van, Network Analysis, Prentice-Hall of India Private Limited (2007).
4. Gayakwad, A. Op-Amps and Linear Integrated Circuits, Prentice-Hall of India (2006).

Reference Books:

1. Chakarbarti, A., Circuit Theory, Dhanpat Rai and Co. (P) Ltd. (2006).
2. Roy Chowdhury, D., Networks and Systems, New Age International (P) Limited, Publishers (2007).

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST) and End Semester Examination (ESE)	70
Sessional may include assignments, regular lab assessments, quizzes, and minor projects (report, presentation etc.)	30